

TOC Becoming Industry Standard for Continuous COD Monitoring

Metrohm's Low Cost Continuous TOC Analyser Shows Benefits of Real Time COD Analysis



Introduction

A Major UK Soft Drinks manufacturer has instantly seen the benefits of moving to on-line technology. The customer approached Metrohm Process Analytics, after their effluent plant (containing an aerobic treatment facility), was overloaded with COD (organic load) over a two day period. The consequences of such an incident were severe. Too high COD load leads to shock and death of the bacteria. The effluent plant had to be closed down whilst the reactors were re-seeded with aerobic bacteria. But whilst the effluent plant was shut, the plant needed to continue production of soft drinks. This meant that effluent was tankered off site until the effluent plant came back on-line. Re-seeding and tankering waste came at a cost of over £0.5million pounds to the plant.

The Problem

The effluent plant was monitoring load with twice daily spot samples using the standard COD lab method. Not only is the test method hazardous with the use of potassium dichromate, it is a time consuming and slow analysis. This particular incident also highlights how spot sampling is not an effective means to monitor or control a process, as significant events can be missed leading to monetary losses and/or out of consent discharges.

Real-time Solution

Metrohm's solution was to monitor and control the daily load to their effluent treatment plant by using Metrohm's continuous TOC analyser. COD is essentially an indirect measure of Organic load (TOC). TOC can be used to measure COD levels with a

correlation factor and it is this method that has become the industry standard for continuous COD monitoring and control.

It was established that in soft drinks manufacture the main contribution to COD was sugars, which would give a consistent chemistry going to the effluent plant and would ensure a consistent, stable correlation between COD and TOC.

Results

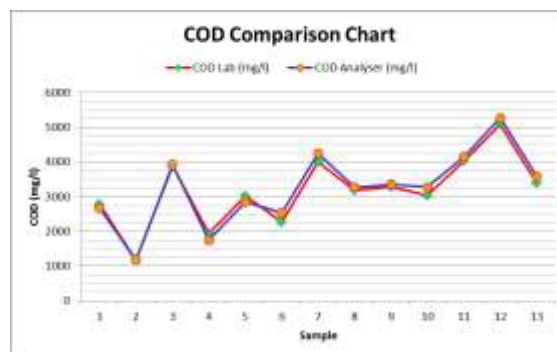
A two month trial period using the TOC analyser was carried out where the company measured lab COD samples alongside the TOC results.



TOC Analyser installed at major UK Soft Drinks Manufacturer on influent sump to treatment tank.

Indeed a very consistent and reliable correlation was established between COD and TOC which turned out to be a correlation ratio between 2.7-2.9. Validation of the ratio using a factor of 2.9 showed the TOC analyser gave highly accurate COD results (for example a TOC value of 1000mg/L would mean a COD of 2900mg/L).

If you can do it in the lab, you can do it on-line



Validation COD test results on spot samples showed very close values to calculated COD results using factor of 2.9.

Robustness to Suspended Solids

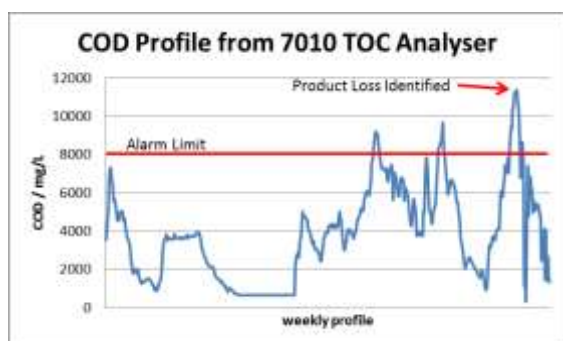
Furthermore the customer was very satisfied that the analyser could cope with the quality of the effluent with a 100% up-time in the face of high levels of suspended solids and bug growth in the sampling lines. This gave them confidence in the robust design of the analyser with wide bore tubing and no internal valves proving the system to be rugged. The simplified layout and use of easily obtainable and inexpensive parts meant the analyser was easy to operate and maintain, whilst giving accurate and reliable COD results continuously.

Instant Return on Investment

The company saw instant benefits when continuously monitoring COD. They were able to understand when peak COD loads were hitting the effluent plant. Also due to the continuous logging of COD they were able to calculate the daily COD loading to the plant and efficiently dose nutrients and save energy costs on aeration to the plant to maintain the correct balance for an efficient treatment. In doing so it has been calculated that the cost of the analyser was paid back within 6 months from chemical savings alone.

Up-Stream Process Understanding

It quickly became apparent that the plant was often being overloaded at systematic times during the week. The operators were able to manage the loading coming into the plant by instructing when the bottling plant had to stop discharge to the effluent plant in order to manage the correct loading and prevent another costly 'COD shock'.



Continuous Profile of Converted TOC to COD for a Week Period

Additional Cost Savings

An additional benefit since the TOC analyser was installed was that it identified a major product loss when a syrup pipe leaked. This was identified by a sudden high level COD spike. The plant was informed; the leak was identified and fixed before more costly losses were incurred (calculated loss before leak being found was twice the cost of the analyser).

Other Applications of Continuous TOC

- Effluent Consent Monitoring and Control
- Monitoring Breakthrough in Boiler Water
- Monitoring Surface Water Discharges
- Monitoring and Control of Coagulation
- Monitoring and Control of Re-cycled Water
- Organic Contamination/Build Up in Processes

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7010 TOC Specification

Analyte	Total organic carbon
Method	Inorganic carbon is removed by acidification and sparging: this is followed by UV-promoted persulfate oxidation. This process oxidizes the total organic carbon into carbon dioxide which is measured in a nondispersive infrared (NDIR) analyser.
Range	0-5mg/L to 20,000 mg/L
Measurement type	Continuous
Lower determination limit	0.2 mg/L
Accuracy	± 2% of full scale nondiluted, ±4% of full scale diluted ranges
Response time	From 6 minutes, depending on range
Ambient temperature	5-40°C
Sample temperature	2-70°C
Sample inlet temperature	Pressureless from overflow vessel
User Interface	Touchscreen
Data logger	Integrated, data download to USB stick
Size	760 x 600 x 210mm
Weight	37kg (approx. depending on range)
Power Supply	115 or 230 VAC 50-60 Hz, 350 VA (115 VAC), 250 VA (230 VAC)
Carrier gas	Air purifier integrated, supplied by an internal compressor
Reagents	Phosphoric acid and sodium persulfate (approximately) 10 L/month for continuous operation)
Analog Outputs	2 x 4-20 mA outputs for measured data
Alarms	2 SPDT contacts. Relay A is programmable - online, offline, loss of sample, result alarm, validation alarm, reagent alarm, calibration alarm, calibration alarm, Relay B is for the instrument fault alarm
Extra relay	Programmable for external operations
Digital input	Remote start/stop
Autoclean, autocalibration, autovalidation functions	Can be selected using the dedicated peristaltic pump
Dual channel	Dual channel integrated
Dual range (low/high)	Switches sample to an external dilutor for a higher range once a set-passed
Factor	Result multiplication factor, e.g. for converting TOC to equivalent COD value
Protection class	IP54
Conformity	EN 610004-2, EN 610004-4, C46-022, EN 55022, EN 61326 (electromagnetic compatibility)